

CDMA通信

假设有ABCD4个站在进行通信，收到的码片序列为S，那么任一站与S的向量内积为：

$$\begin{aligned} A \bullet S &= A \bullet (A+B+C+D) \\ &= A \bullet A + A \bullet B + A \bullet C + A \bullet D \\ &= A \bullet A + 0 + 0 + 0 \\ &= \begin{cases} A \bullet A = 1 & \text{发送原码} \\ A \bullet 0 = 0 & \text{未发} \\ A \bullet \bar{A} = -1 & \text{发送反码} \end{cases} \end{aligned}$$